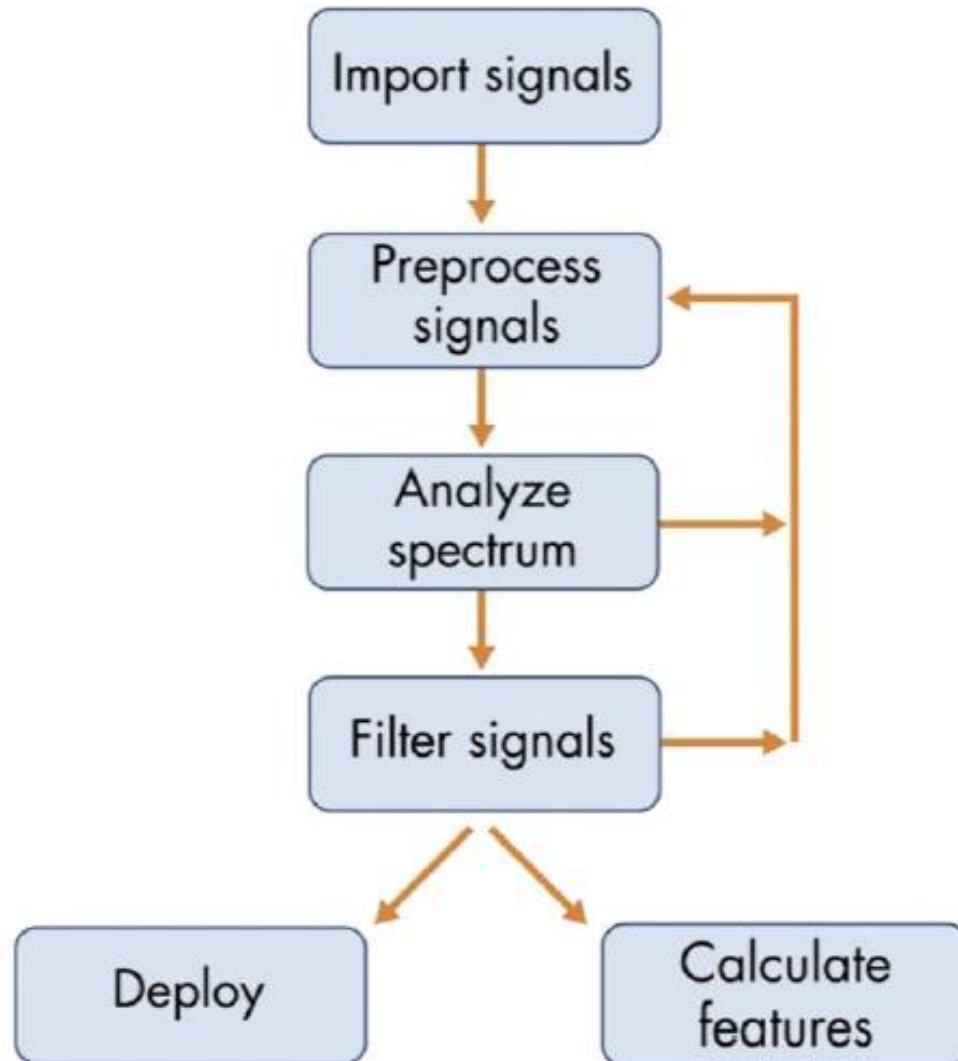


Signal Processing Using MATLAB

18th Feb 2024

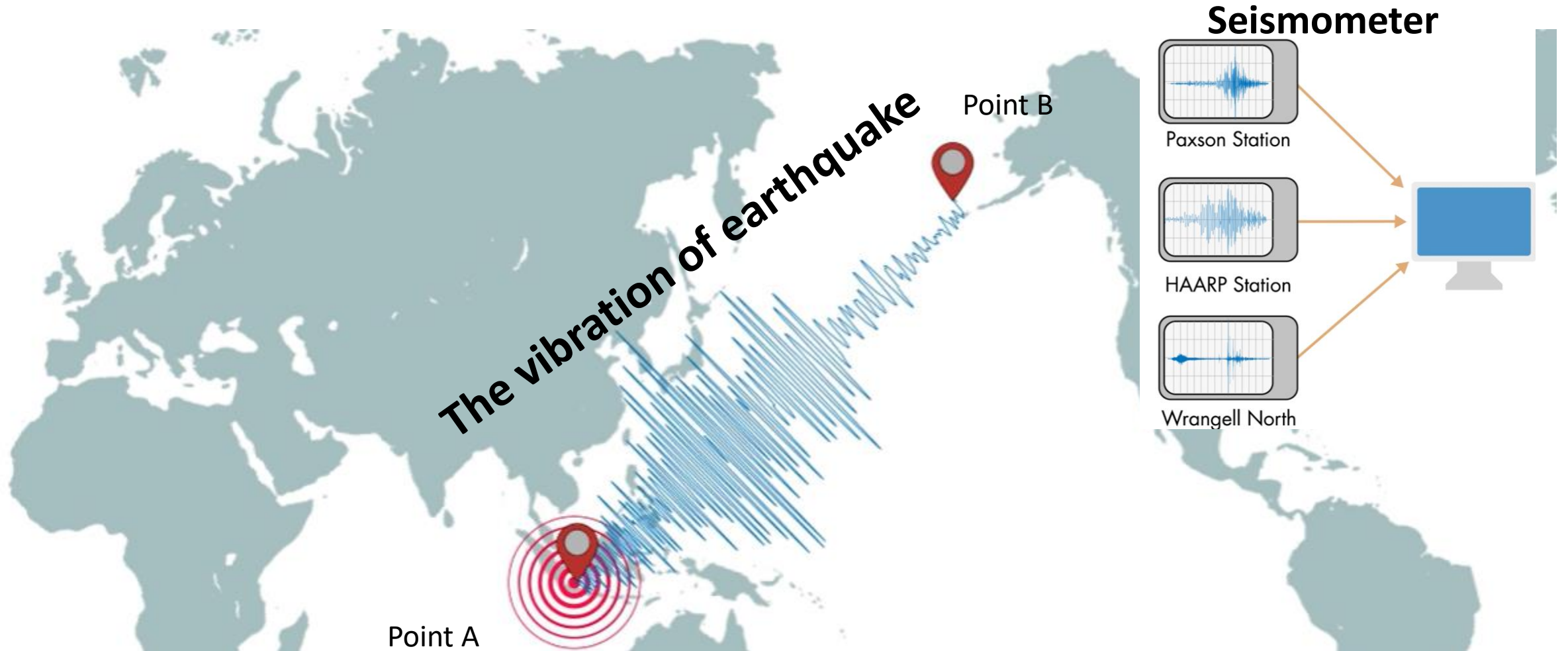
Dr. Saif Ali Jasim Al Grait

Overview: Signal Processing Workflow

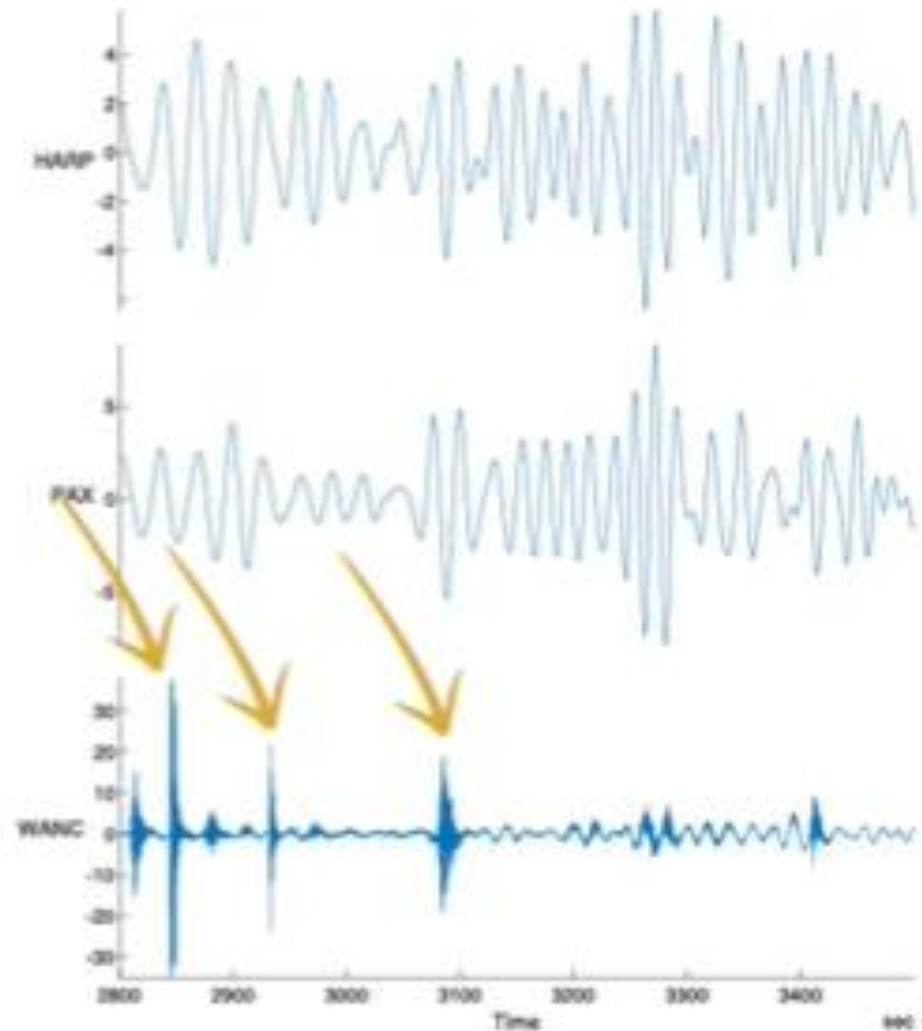
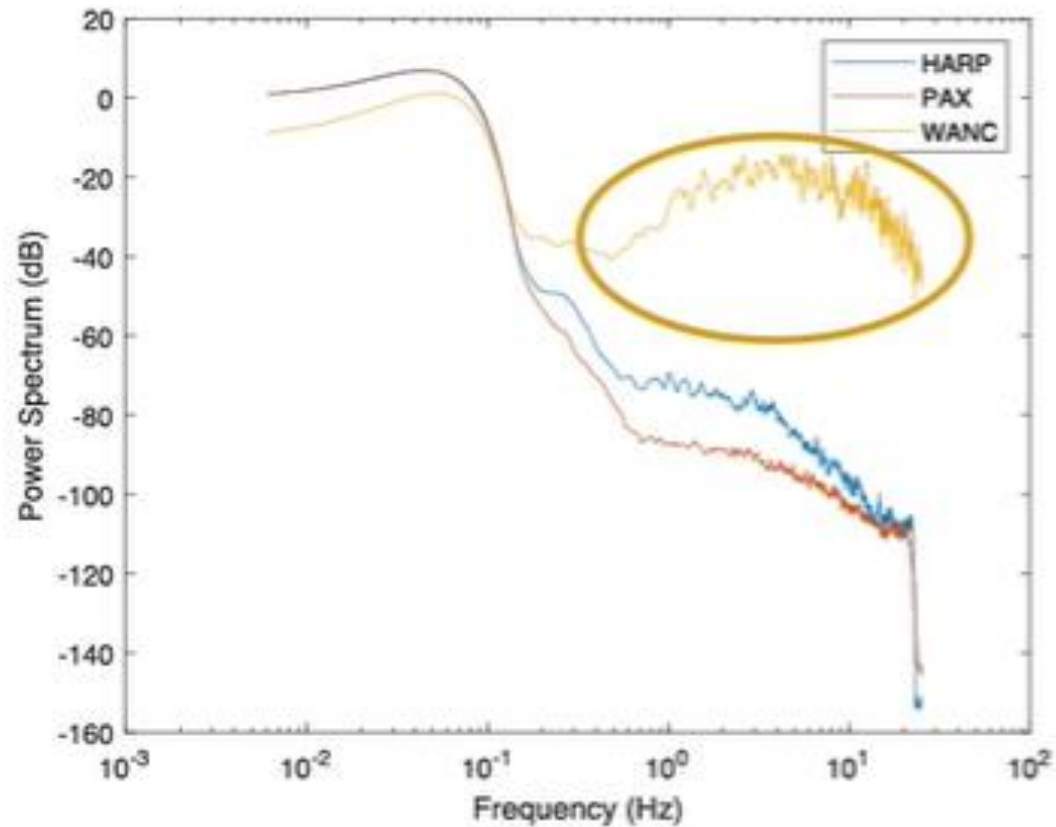


1. Import the signals into MATLAB
2. Preprocess the signals in the time domain to prepare them for analysis
3. Look at the power spectrum in the frequency domain
4. Filter the signals to keep only relevant frequency content
5. Revisit earlier steps in the process to refine your algorithm
6. Extract information from the signals or deploy your algorithm to hardware

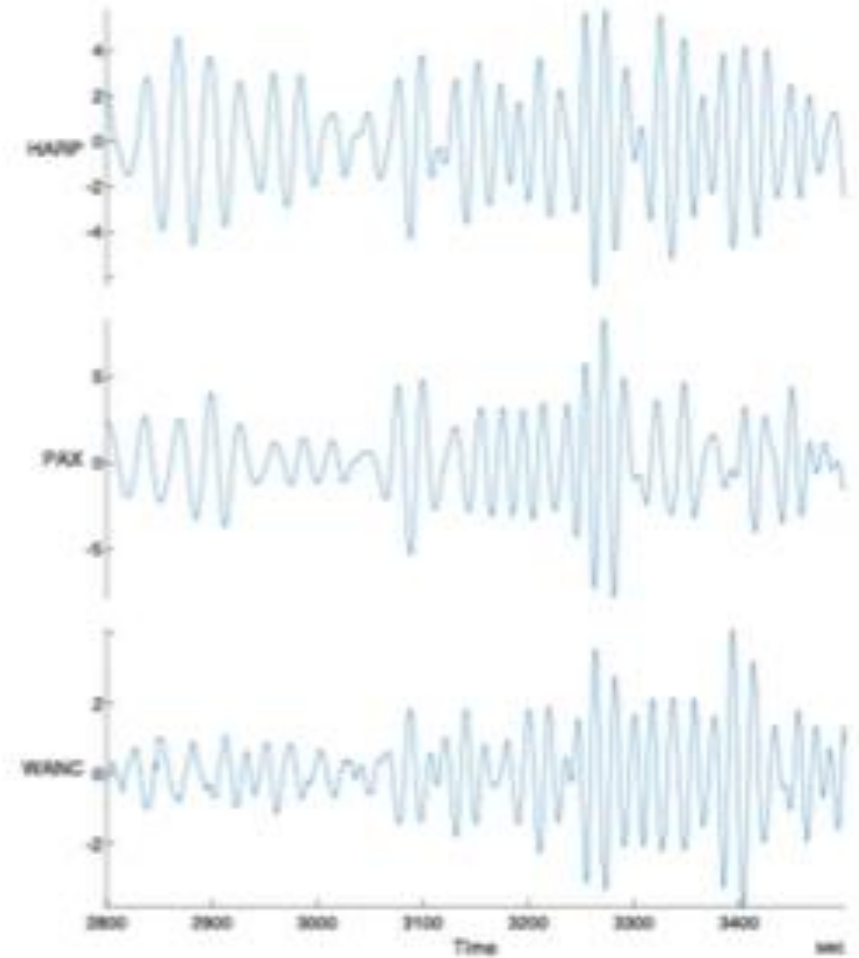
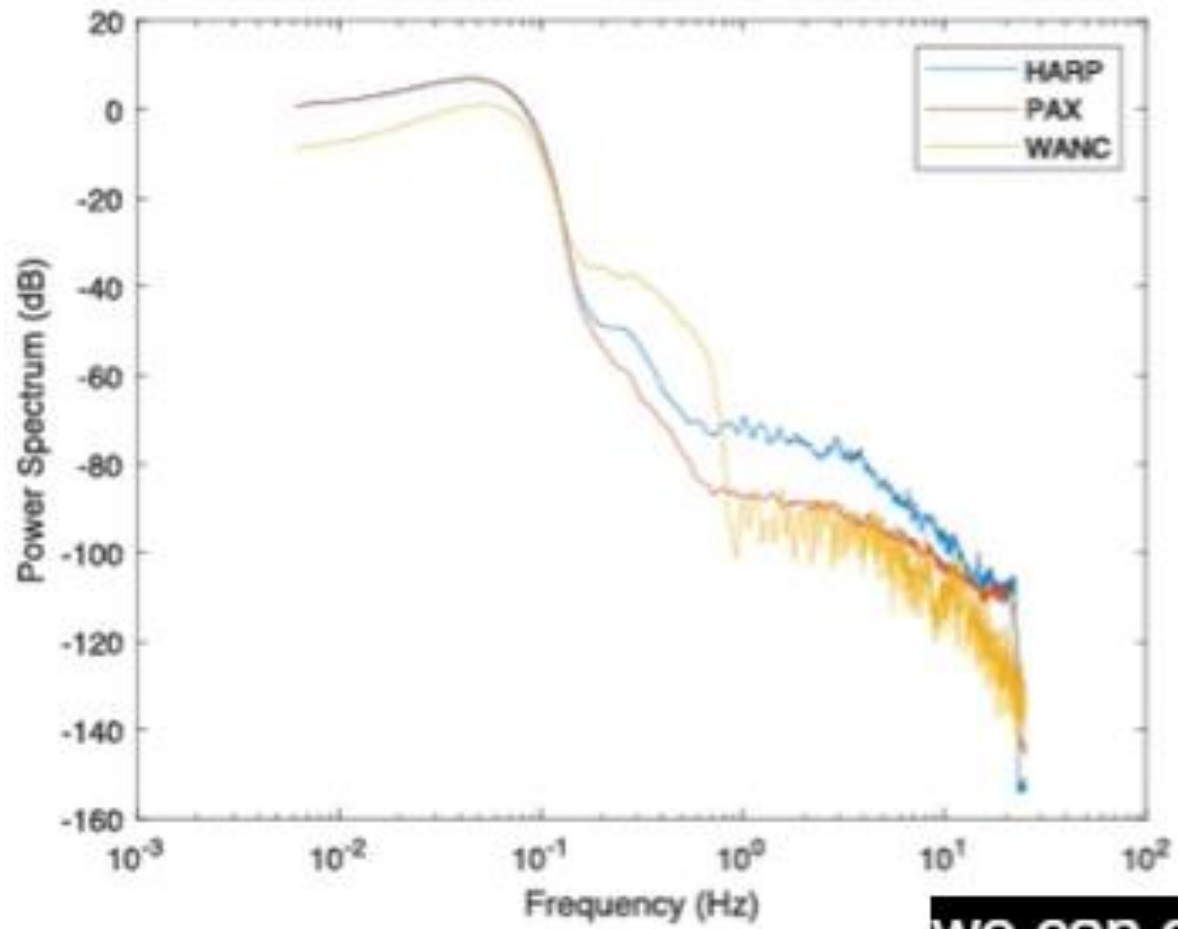
Task 1: Perform signal processing for earthquake



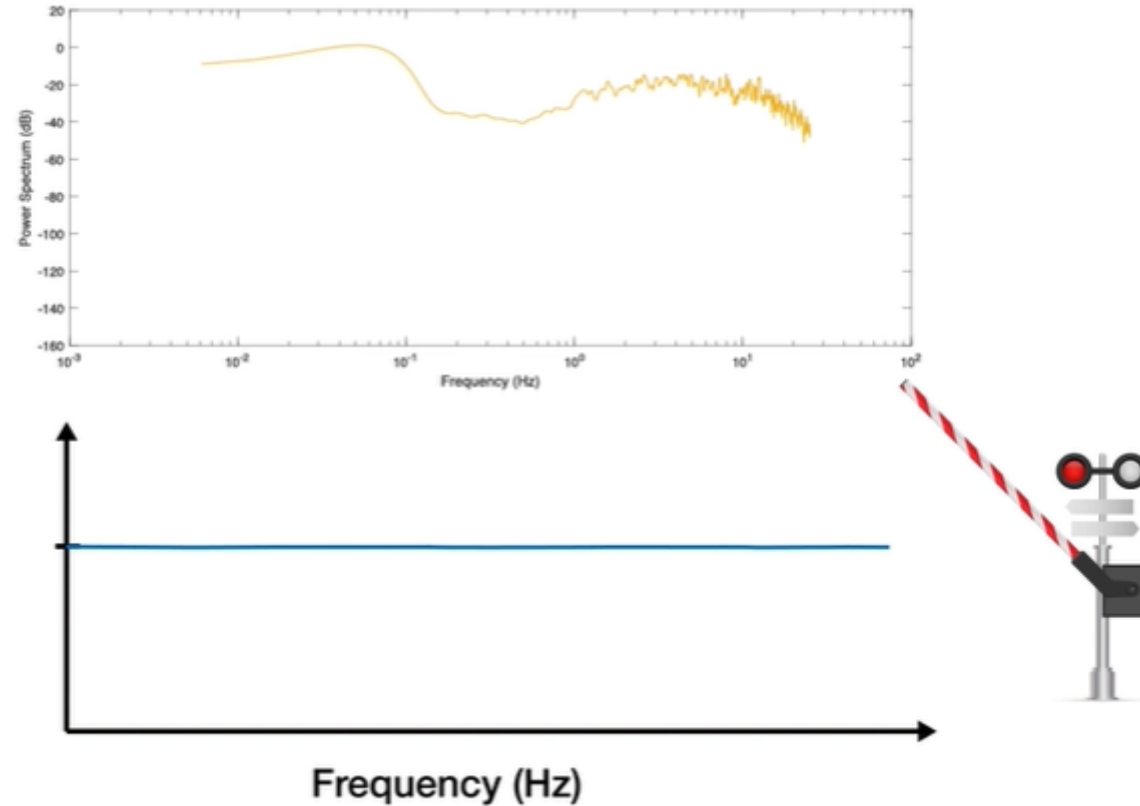
Wrangle signal has high frequency component that makes the signal is different in the time doming



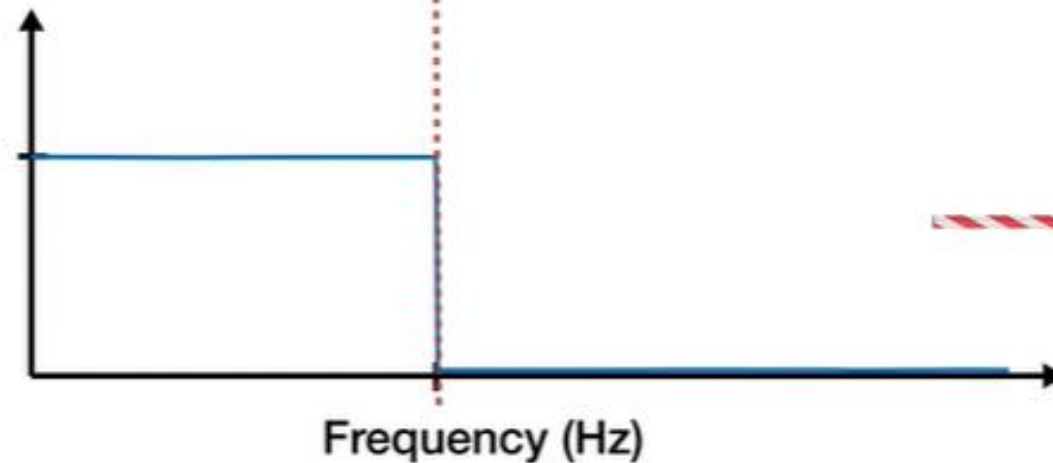
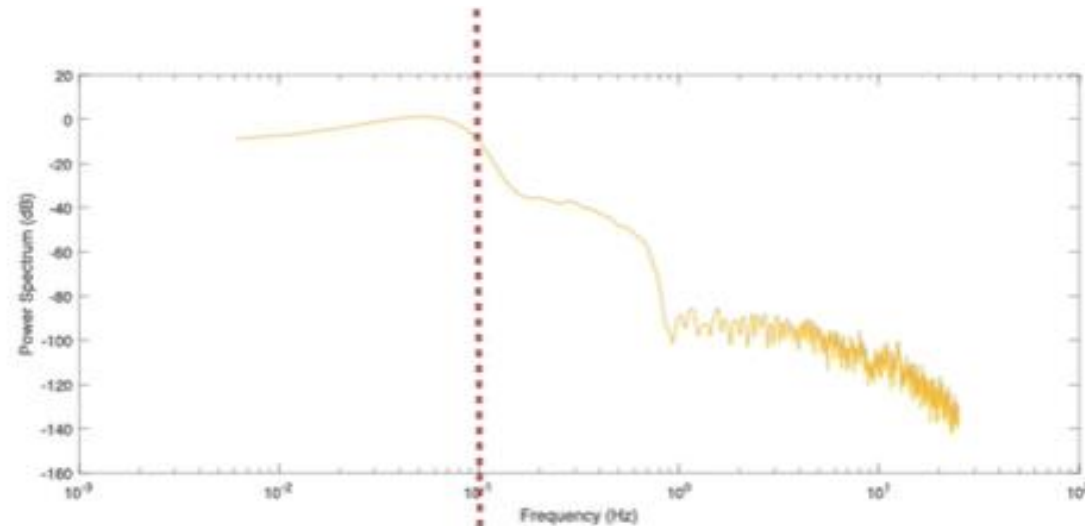
It is hard to see the low-frequency surface waves, by cutting out the high-frequency content to compare all three signals in time domain



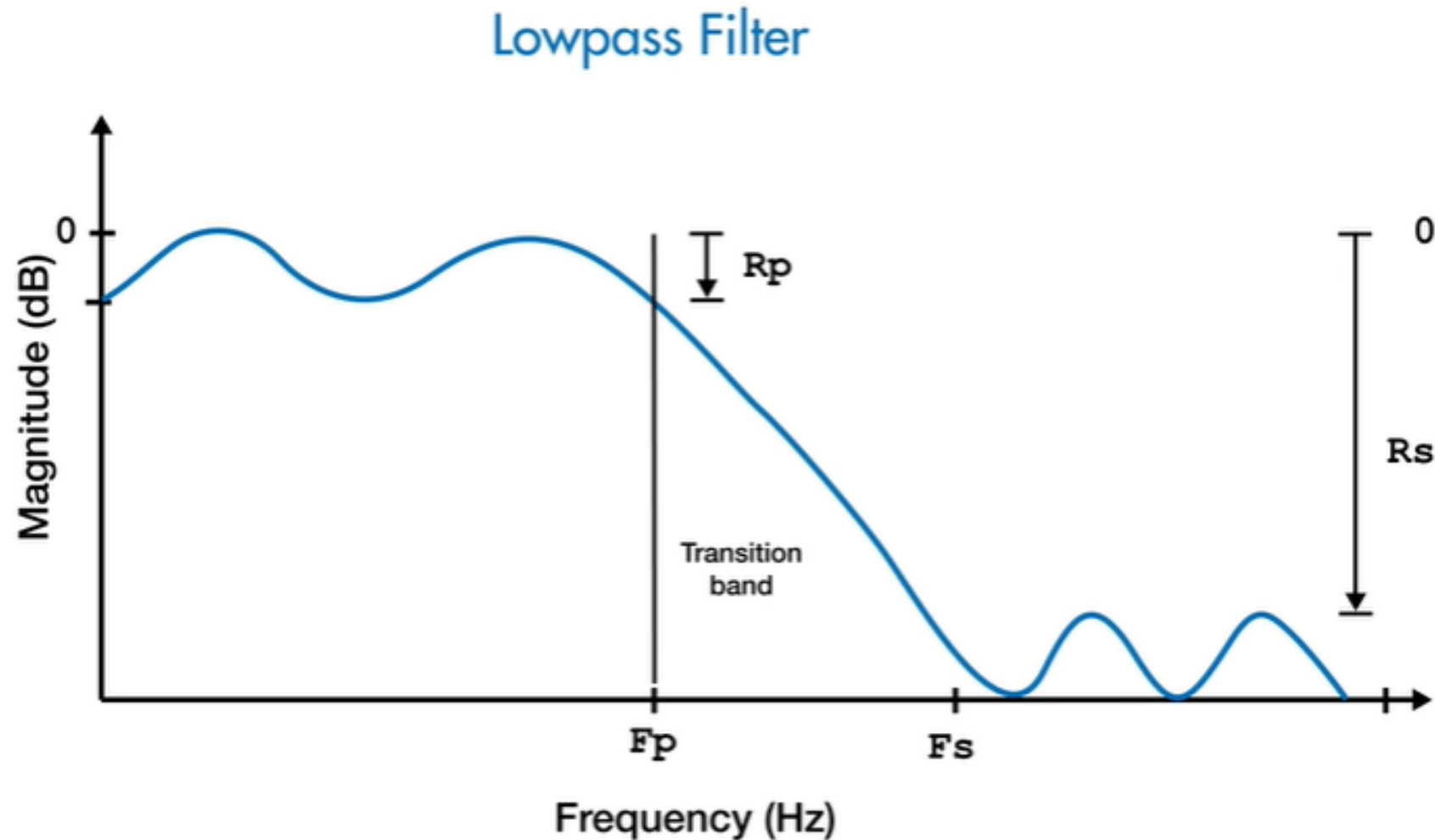
A filter can remove content from a signal based on its frequency



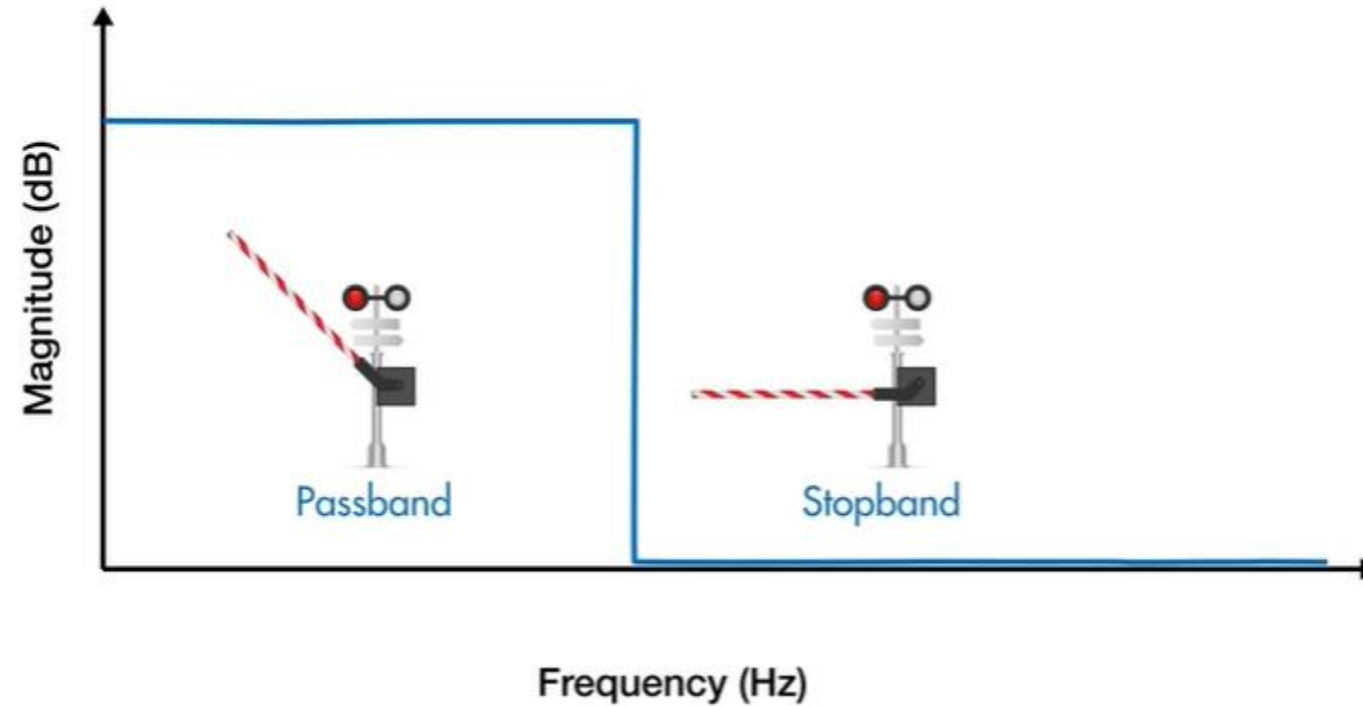
To remove everything but the low frequencies, a low pass filter is needed



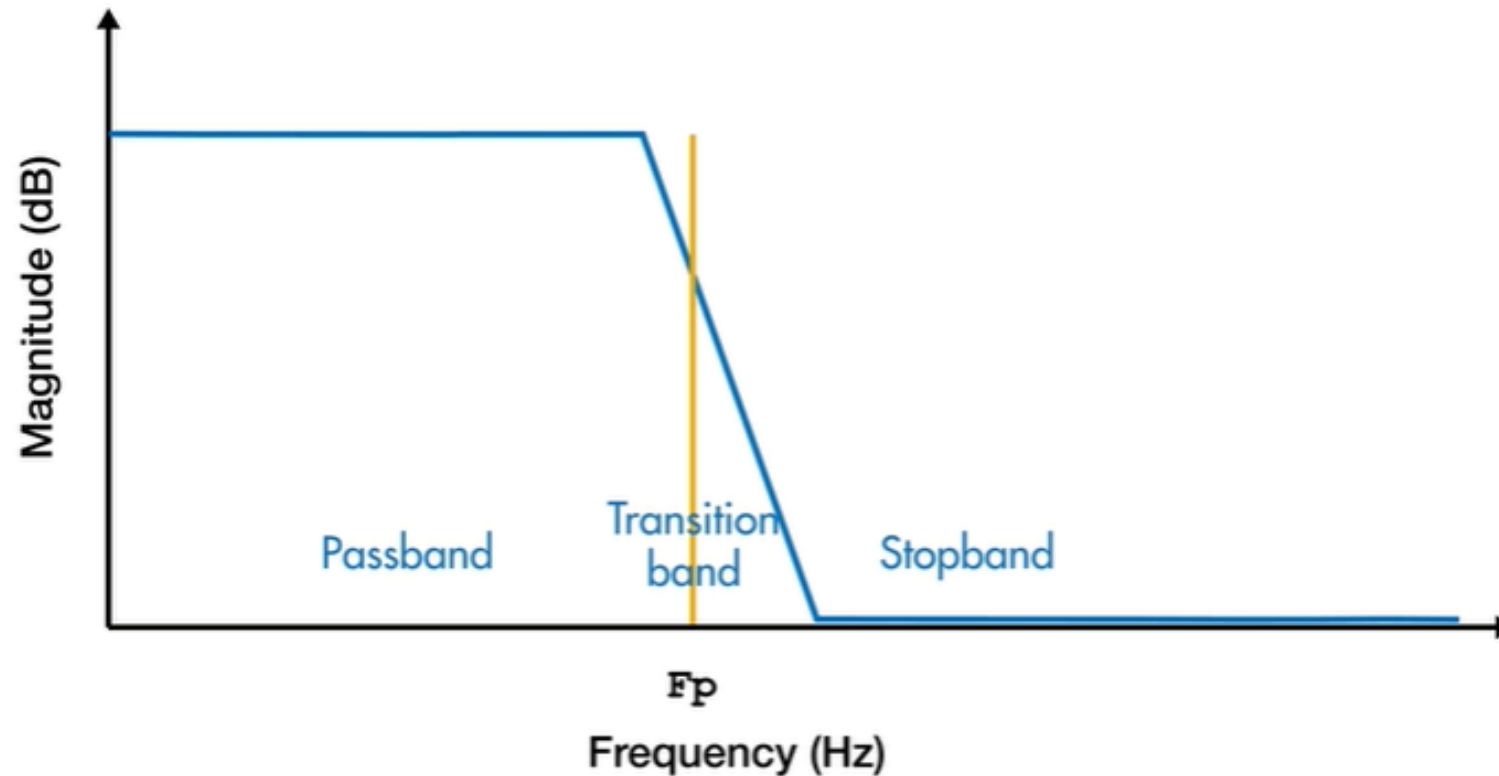
Fitter parameters design can be optimized to enhance performance



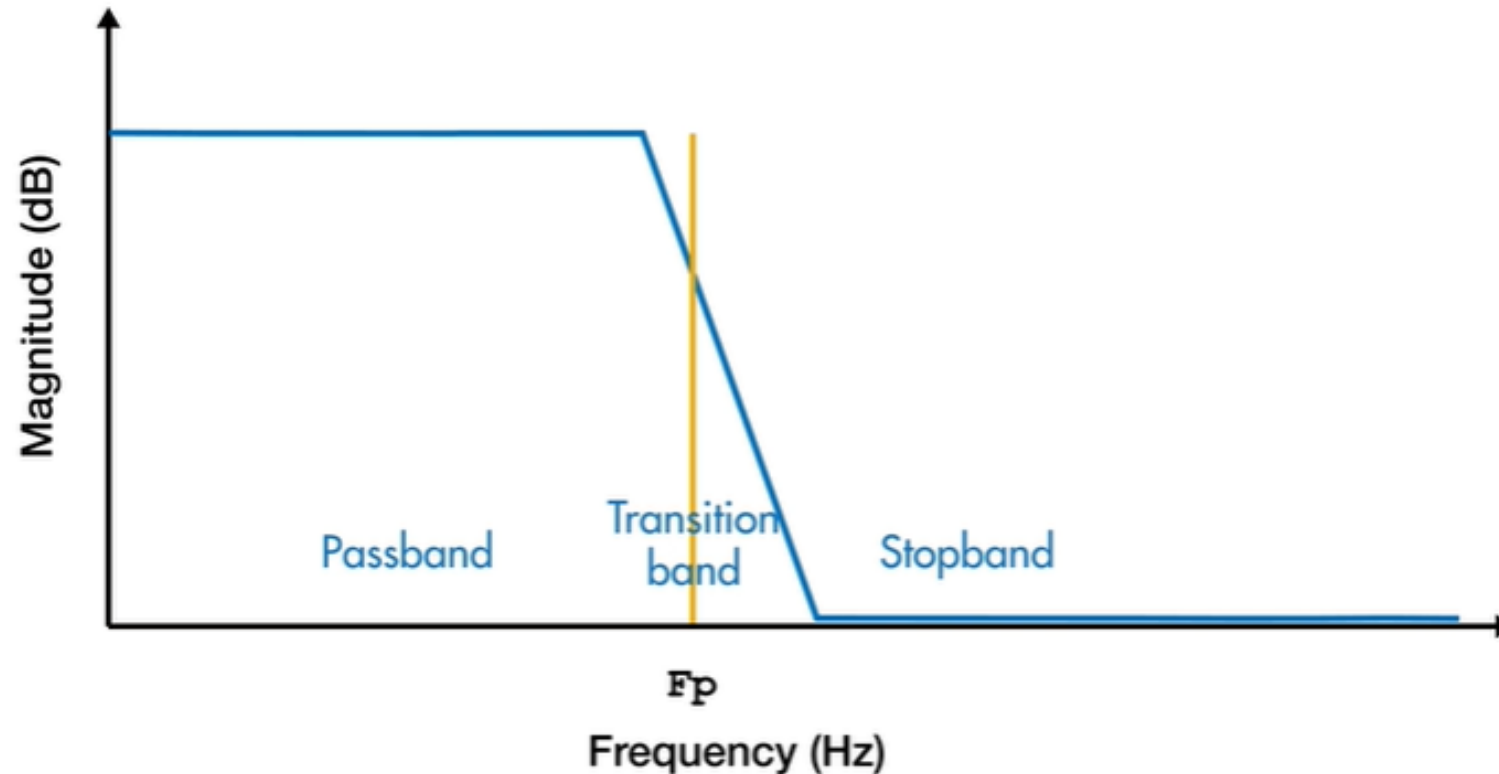
Ideal filter with low frequency pass-band



Ideal filter with low frequency pass-band, but in reality there is a transition band

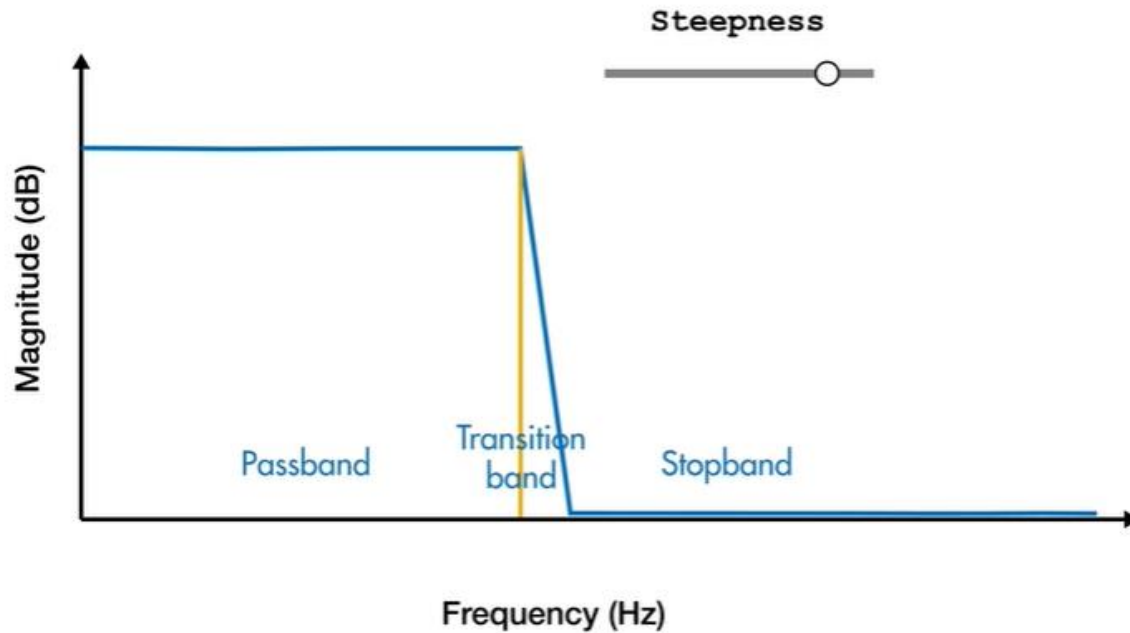


The size of the transition region determines how much unwanted frequency pass through the filter

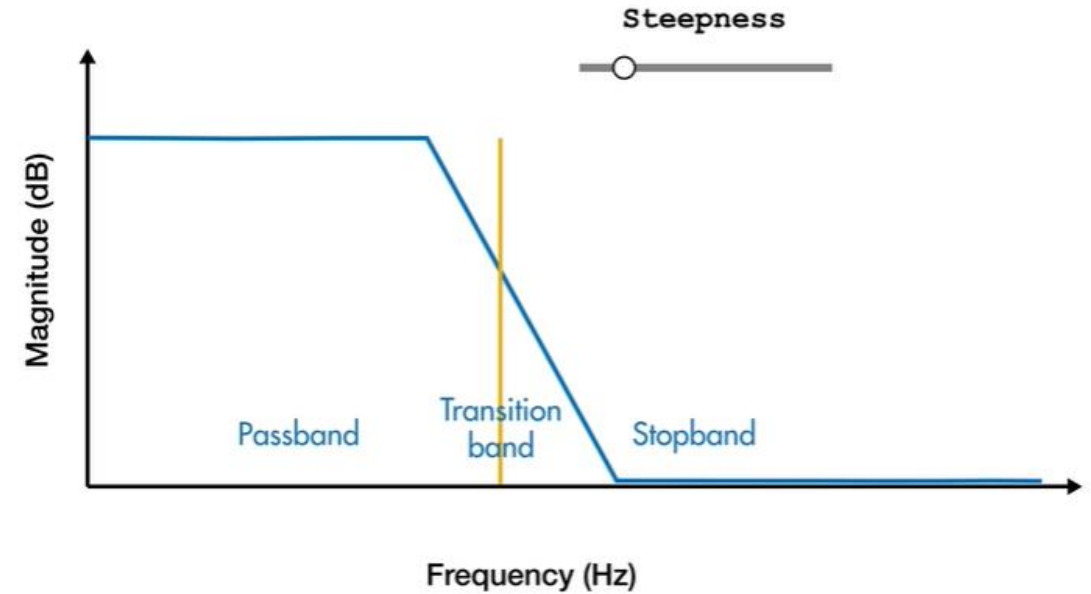


The width of the transition band can be controlled with **steepness** option

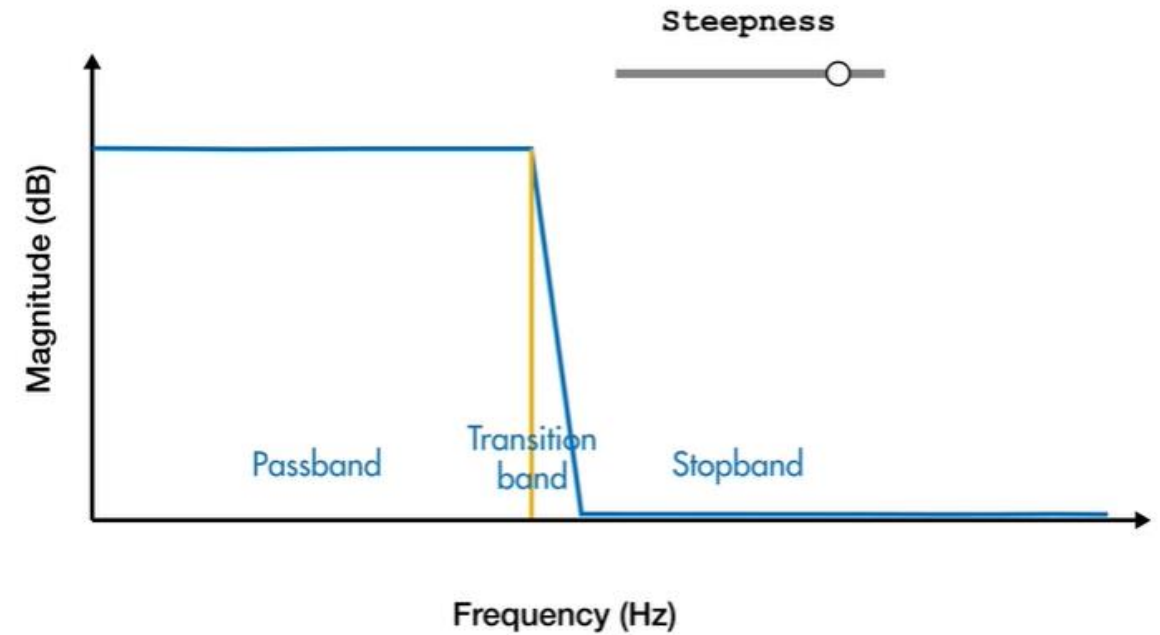
High steepness gives narrow transition band



Low steepness gives wide transition band



Why don't set the steepness to maximum value?

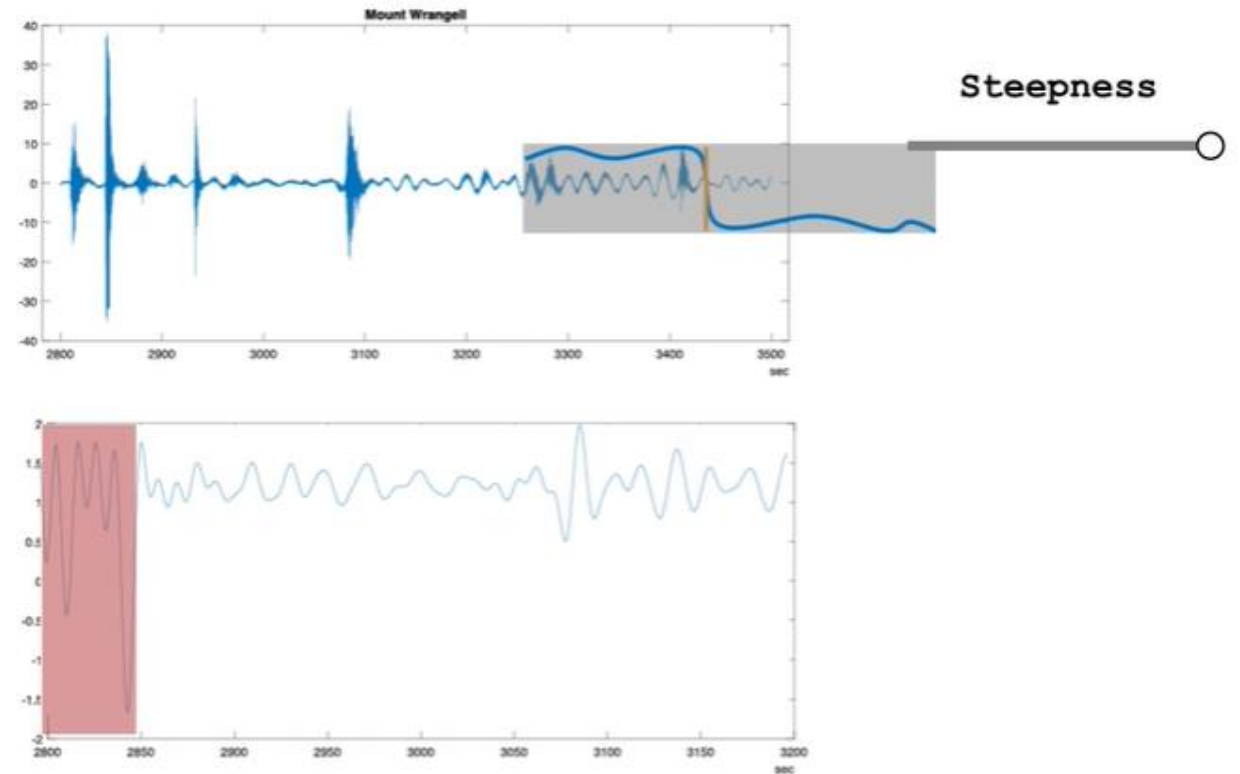


Why don't set the steepness to maximum value?

There are some tradeoffs with filter design because in general filters introduce some artifacts in the beginning and end of signal

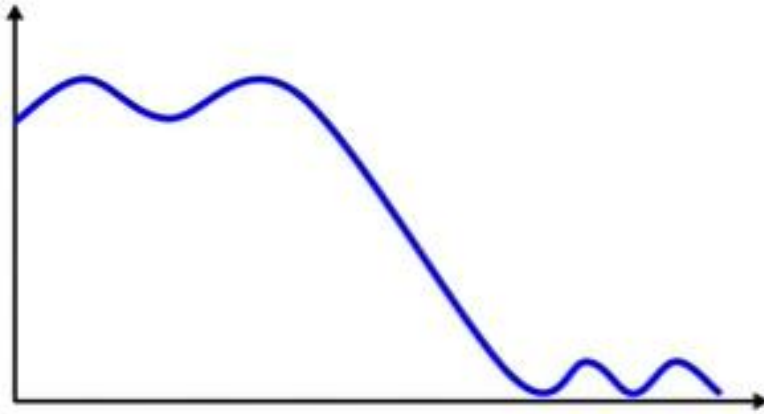
High steepness increases the length of that corrupted region

Try to use the smallest steepness values
That still produce a good output

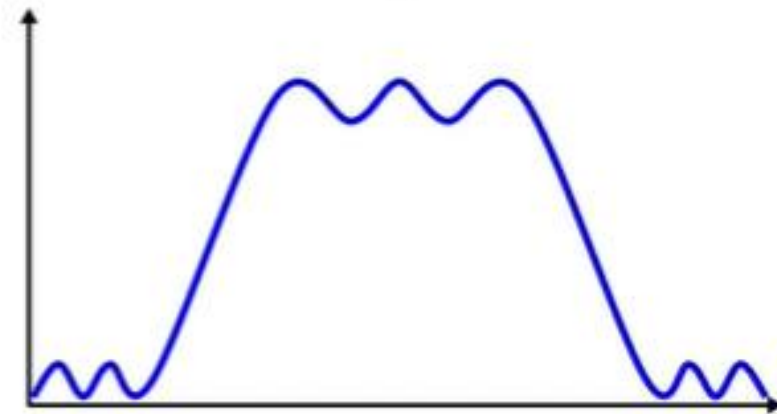


If you need to design other filters the process is similar

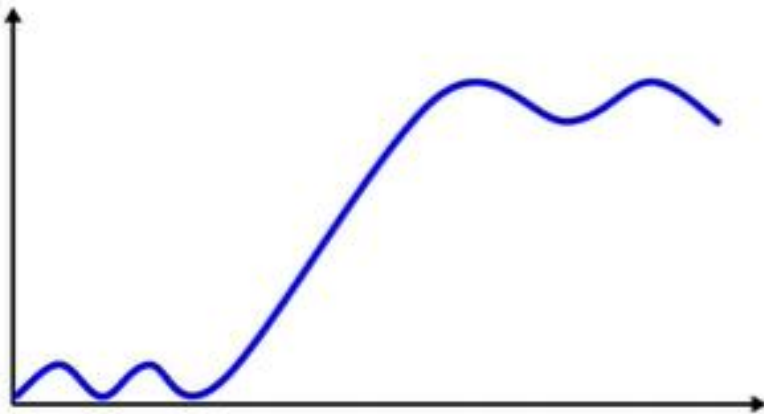
Lowpass



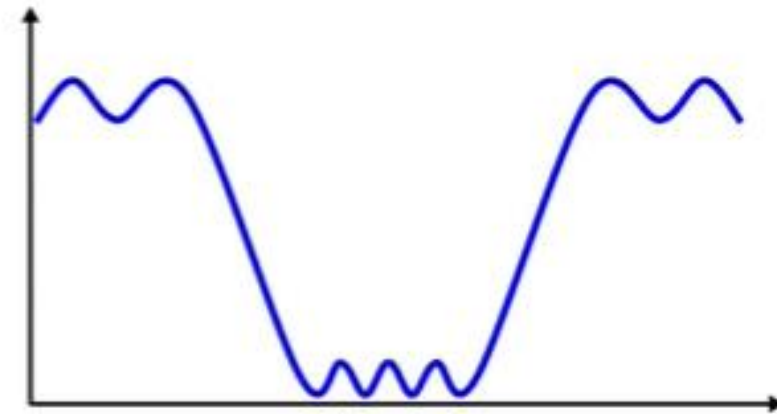
Bandpass



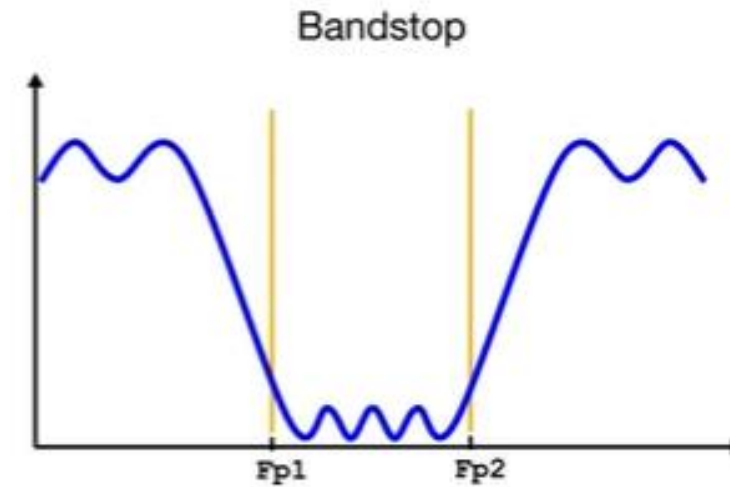
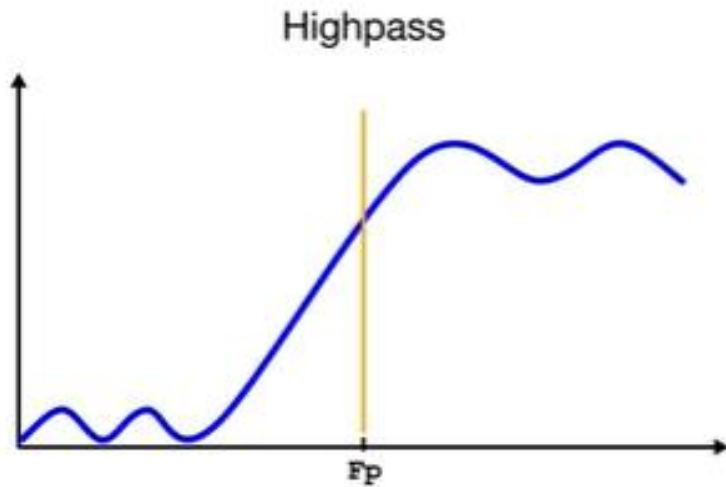
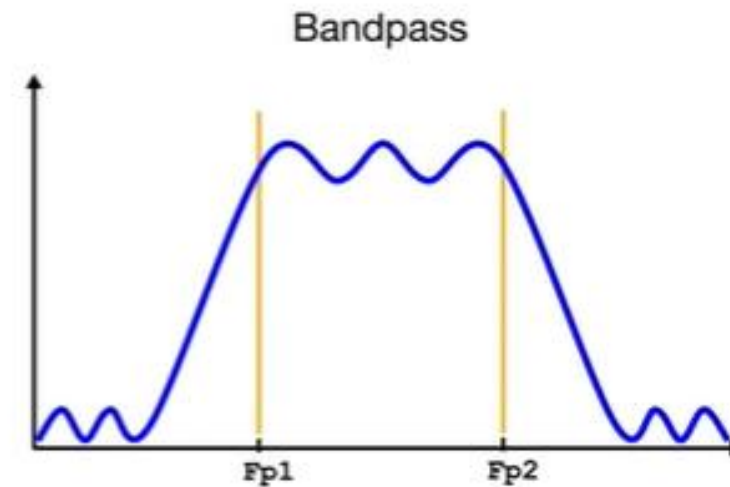
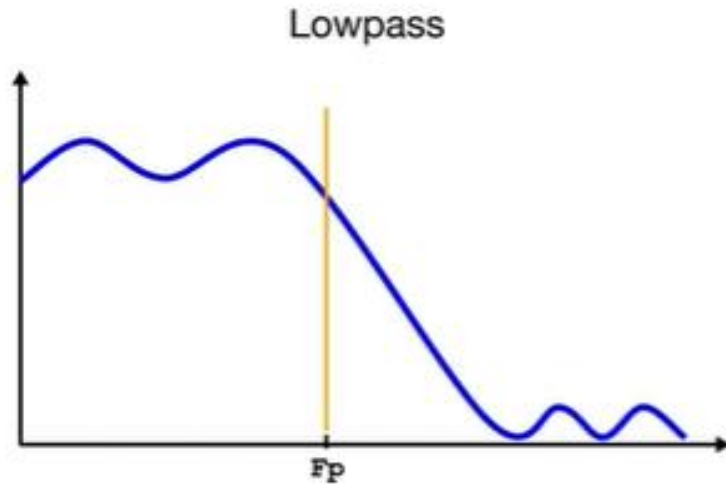
Highpass



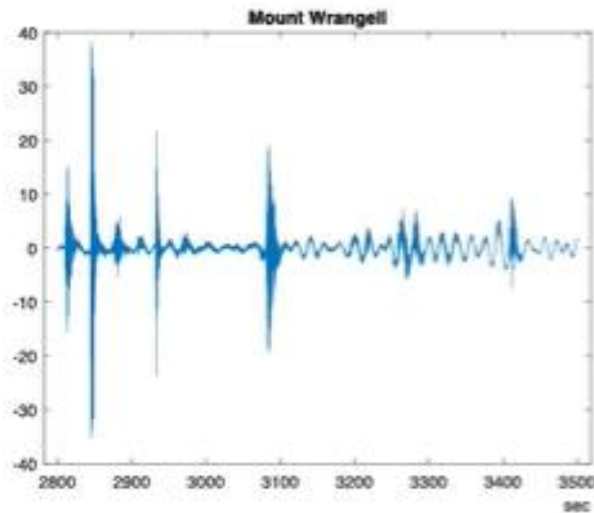
Bandstop



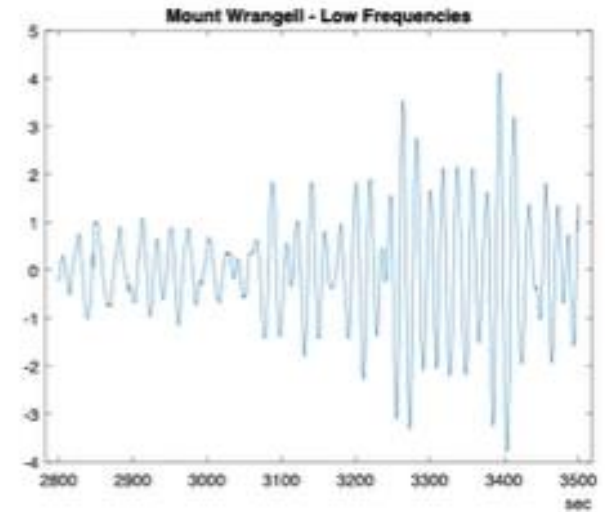
If you need to design other filters the process is similar by choosing the frequency the needed steepness



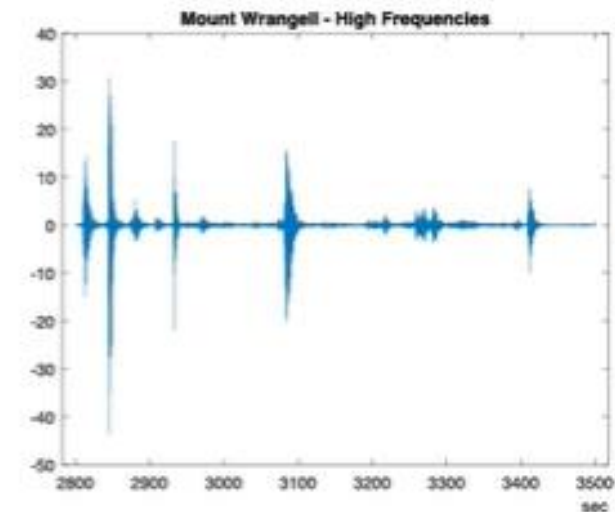
We need to use two filters on the Wrangle signal to separate the surface wave vibration and local earthquake vibration



lowpass



bandpass



Low-pass Filter

The HARP and PAX seismic stations recorded low-frequency surface waves from the Sumatra earthquake. You've seen evidence that the Mount Wrangell seismic station also recorded the same surface waves:

Cross-correlation between the signals while aligning

Power spectra show a similar peak around 0.05 Hz

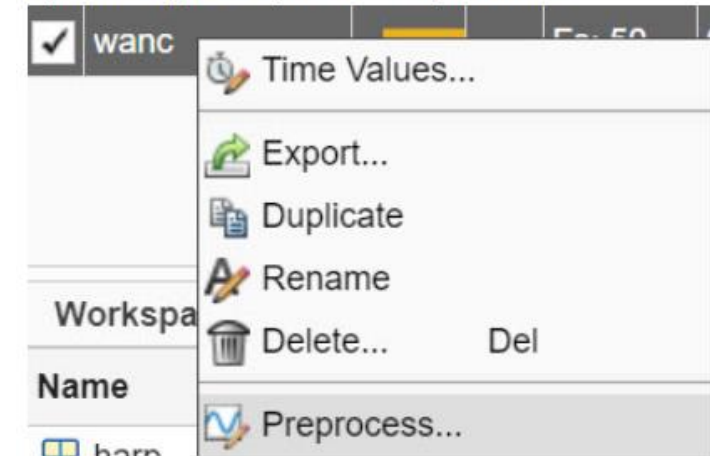
Look at the power spectrum with a logarithmic scale on the frequency axis. The low frequencies are located from about 0 to 0.1 Hz, so you can lowpass filter the wanc signal at 0.1 Hz to see those frequencies in the time domain.

In preprocessing mode, you can find all the filtering options in the Functions dropdown.

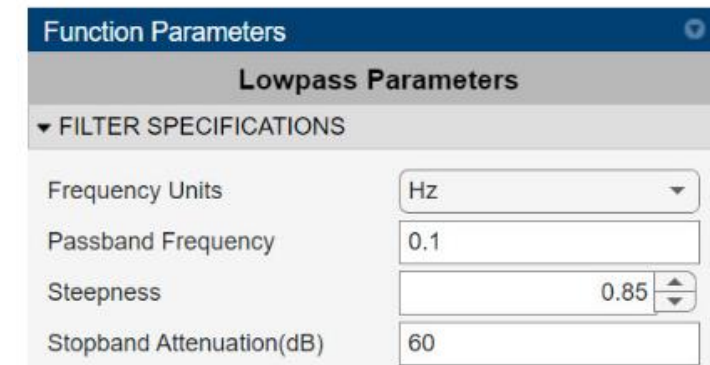
TASK

Duplicate the wanc signal and name the copy wanc_filt.
Lowpass filter wanc_filt at 0.1 Hz.
Then compare the power spectrum of wanc and wanc_filt.

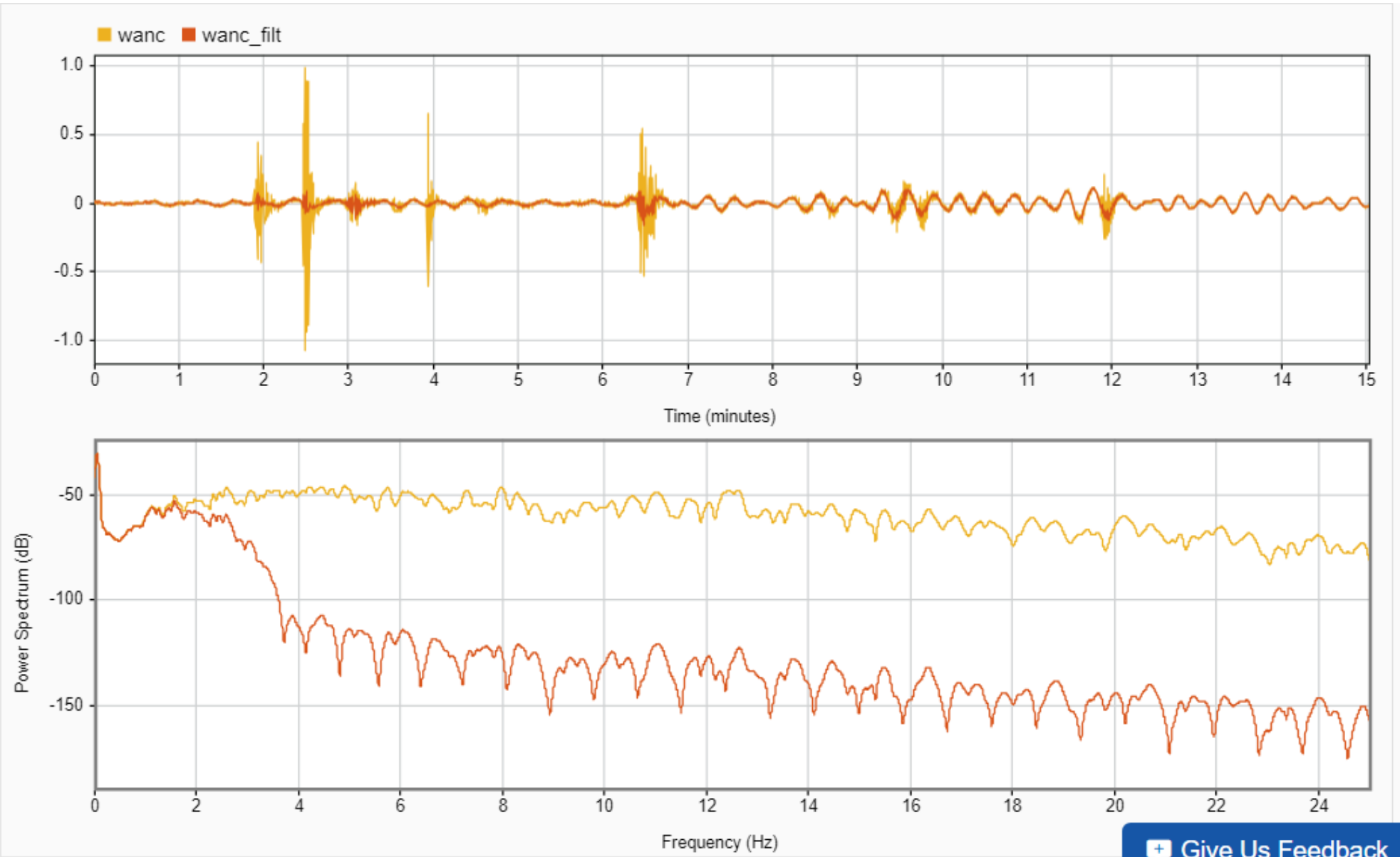
You can find the **Duplicate** and **Preprocess** options by right-clicking on a signal in the Signal table.



In **Preprocess** mode, select **Lowpass** from the **Functions** dropdown. Set the passband frequency to 0.1.

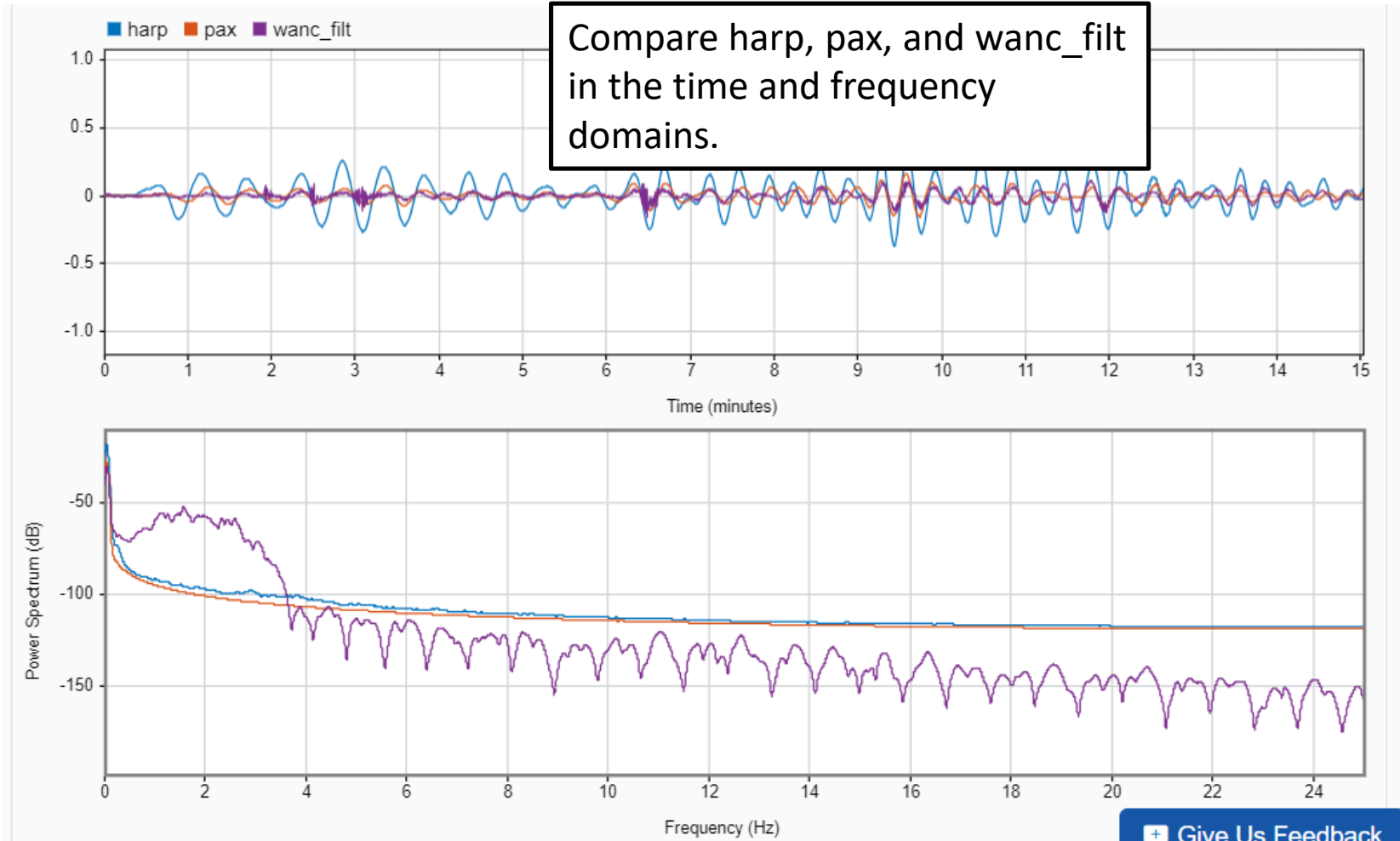


Low-pass Filter



Low-pass Filter: filtering removed high-frequency content from the signal recorded at Mount Wrangell, but how do the low frequencies compare to the signals recorded at the HARP and PAX stations?

In the time-domain plot, you can see that the large spikes were filtered out of wanc_filt. In the frequency-domain plot, there are still frequencies over 0.1, but filtering removed some high-frequency content.



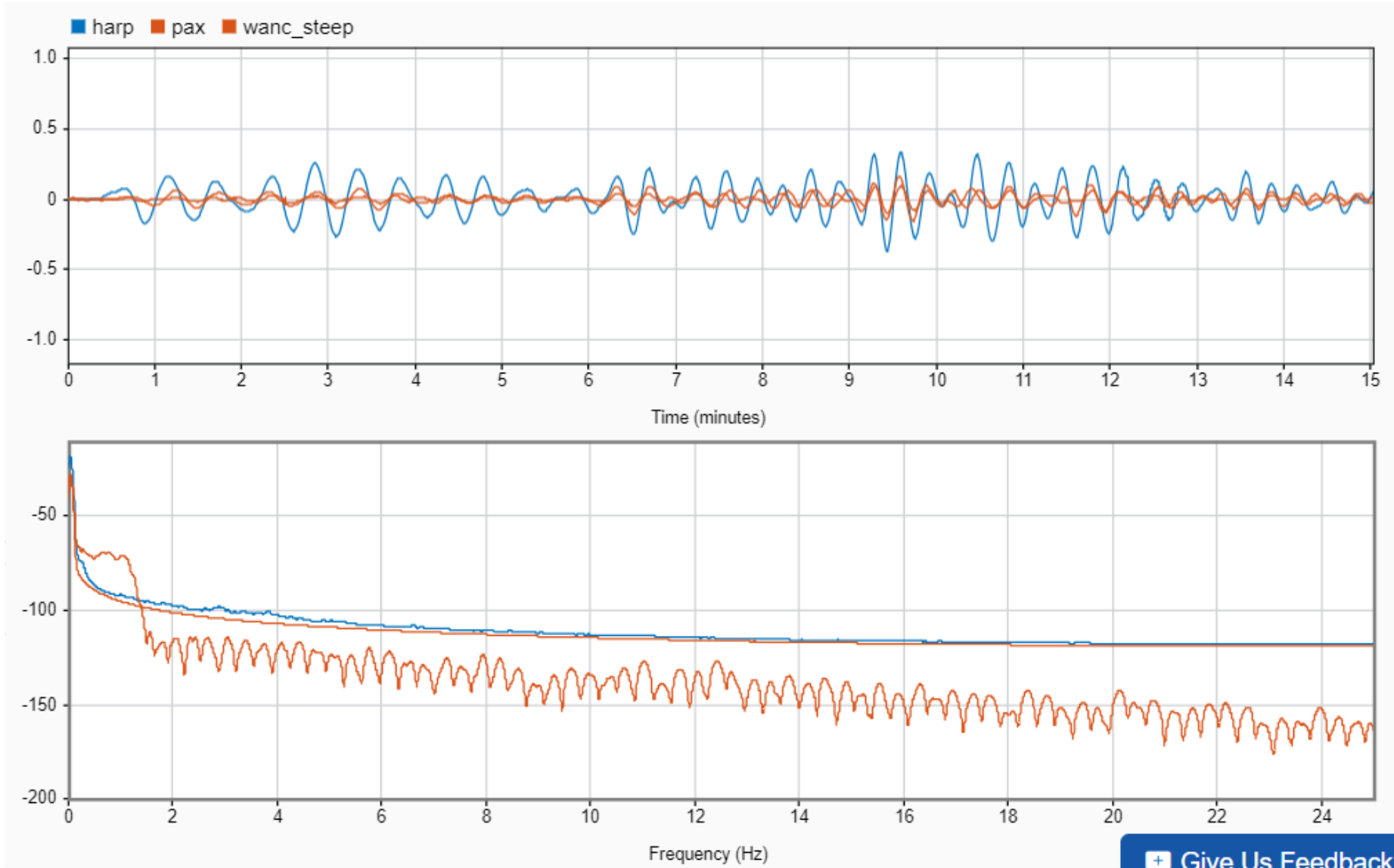
To reduce the high-frequency content in the filtered signal, you can increase the steepness.

The filtered Mount Wrangell signal looks similar to HARP and PAX, but many of the curves are jagged. The jagged curves are from high-frequency content that was left over after filtering.

Create another copy of the wanc signal. Name this copy wanc_steep and lowpass filter it at 0.1 again, but this time, set the filter's steepness to 0.95.

Compare wanc_steep, pax, and harp in the time and frequency domains.

To reduce the high-frequency content in the filtered signal, you can increase the steepness.



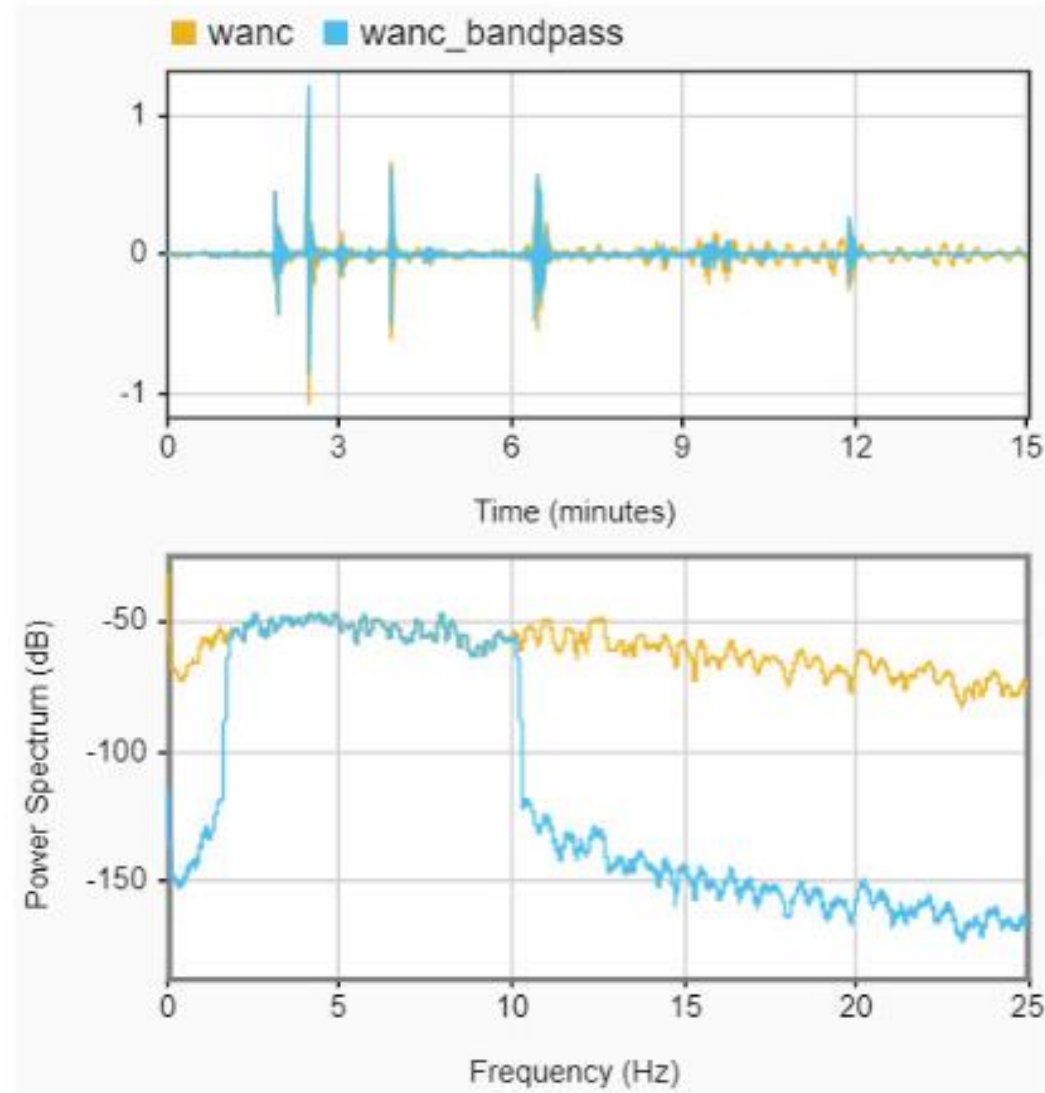
The previous activity, you extracted the low frequencies corresponding to the surface waves from the earthquake in Sumatra

Now you can filter to get the high-frequency content corresponding to the local earthquakes near Mount Wrangell. The frequency range of local earthquakes is 2 Hz to 10 Hz.

To extract these frequencies, you can use a bandpass filter to keep only the frequencies in that range.

Duplicate the wanc signal. Name the copy `wanc_bandpass`. Bandpass filter `wanc_bandpass` from 2 to 10 Hz. Compare `wanc` and `wanc_bandpass` in the time and frequency domains.

The previous activity, you extracted the low frequencies corresponding to the surface waves from the earthquake in Sumatra



Signal Measurements

The spectrogram of the Mount Wrangell signal is displayed. The earthquakes occurred wherever there's a lot of frequency power, so can you use this spectrogram to find the timestamps of the earthquakes?

To compute anything from the spectrogram, first you need to get the arrays instead of just the image.

```
[p,f,t] = pspectrum(sig,fs,"spectrogram")
```

The three outputs are the spectral estimates, the frequencies, and the time instants of each segment.

Edit the code used to create the spectrogram. Add three output arguments named p, f, and t.

```
[p,f,t] = pspectrum(wanc_bandpass,50,"spectrogram");
```

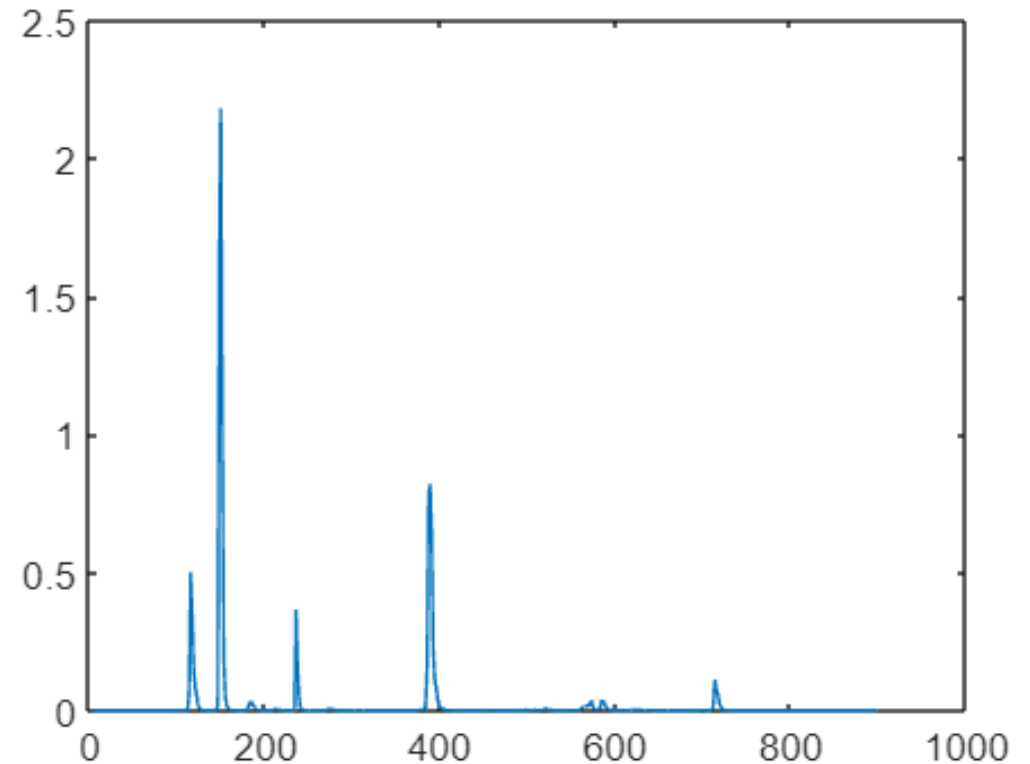
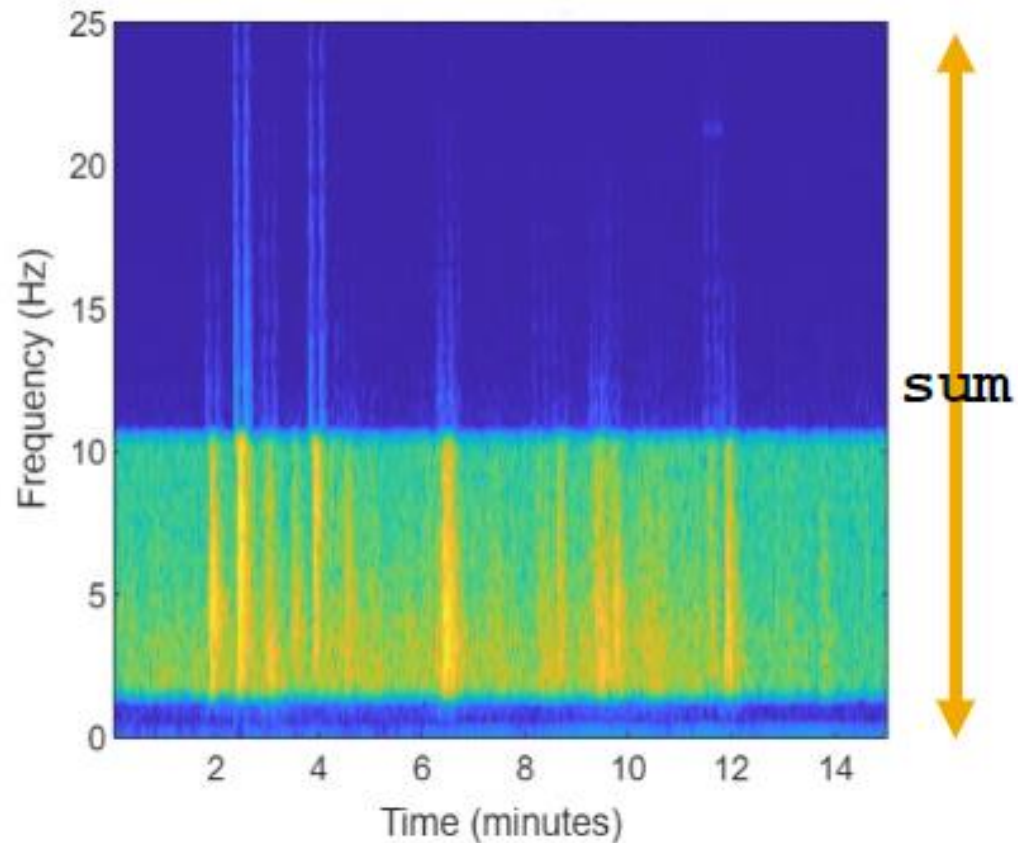
pspectrum

Analyze signals in the frequency and time-frequency domains

<https://www.mathworks.com/help/signal/ref/pspectrum.html>

Calculate the sum of p. Name the output psum. Then create a plot of psum versus t.

To find which timestamps contain a lot of high frequencies, you can sum the power across all frequencies for each timestamp.



Calculate the power of psum and name it pwr. Then create a plot of pwr versus t.

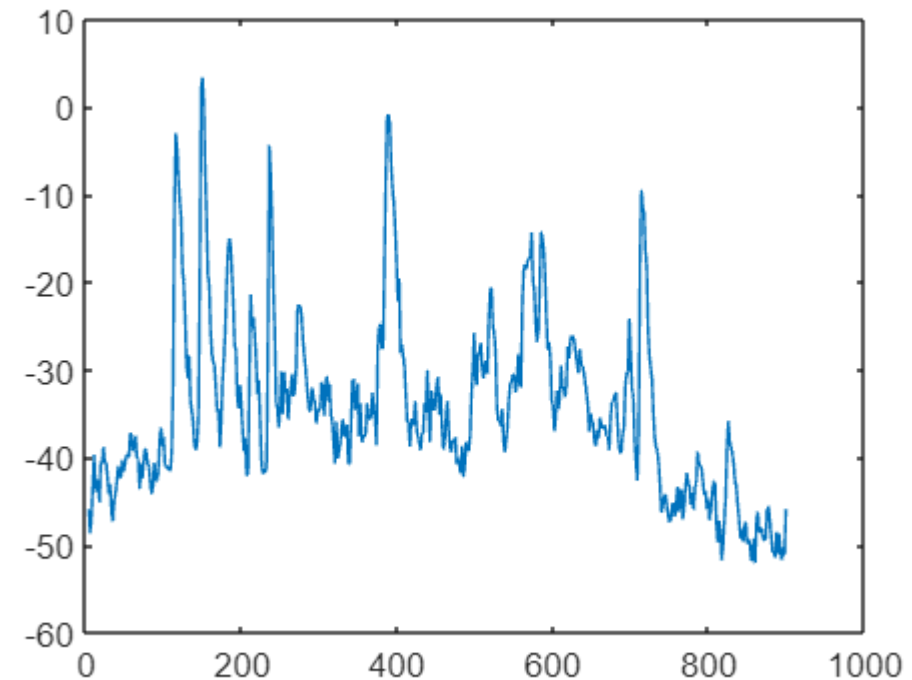
You can see a few peaks in psum, but some peaks are much less prominent than others. Recall that in Signal Analyzer, the spectrum display uses decibels (dB) on the frequency axis. To accentuate the peaks, you can calculate the power of psum.

```
xpwr = db(x,"power")
```

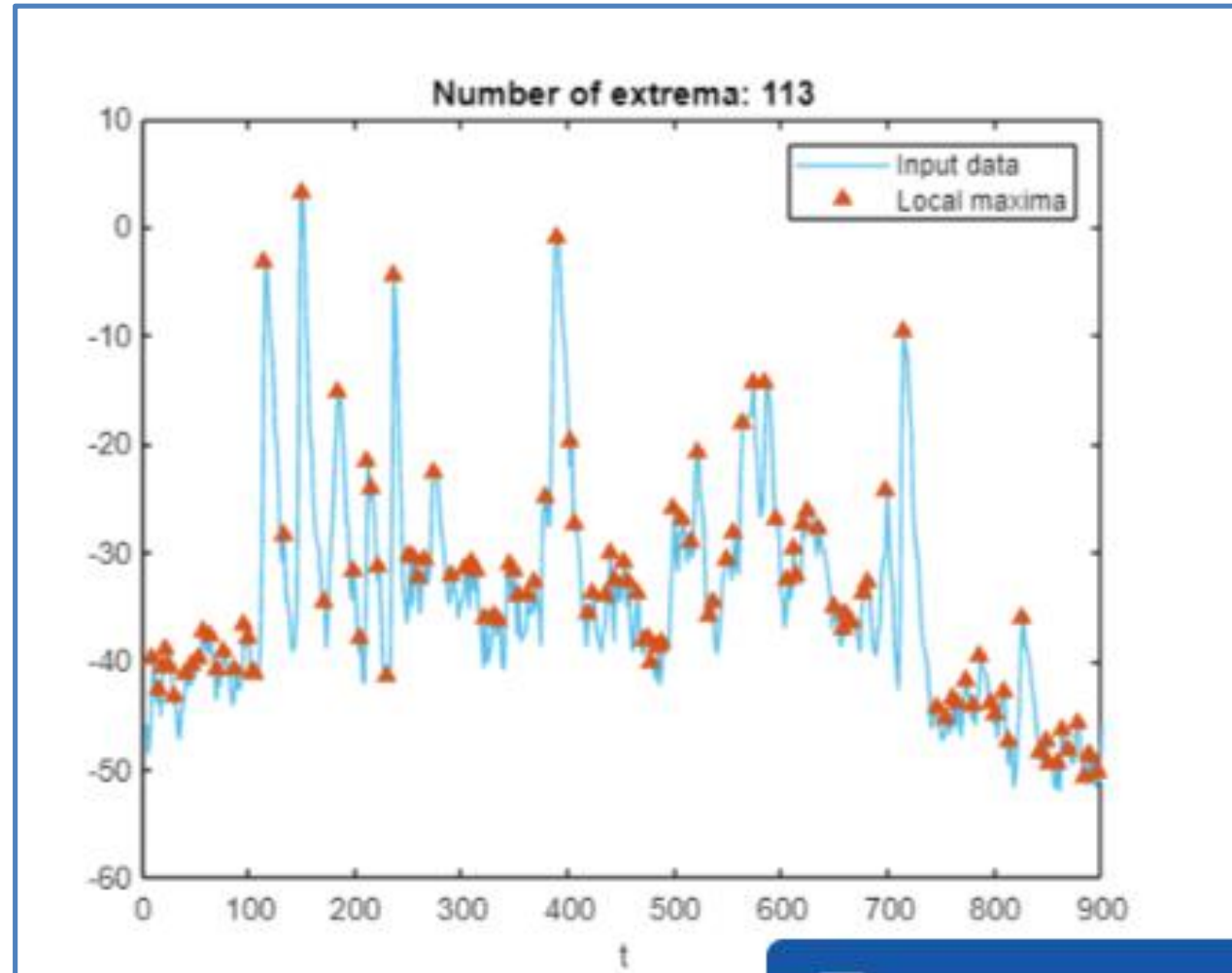
db

Convert energy or power measurements to decibels

<https://www.mathworks.com/help/signal/ref/db.html>



To find the timestamps of each earthquake, you can find the location of these spikes. You can use the Find Local Extrema interactive task to automatically identify local minima or maxima in a signal.



Final Local Extrema using live editor

Find Local Extrema

Find local maxima and minima in the Live Editor

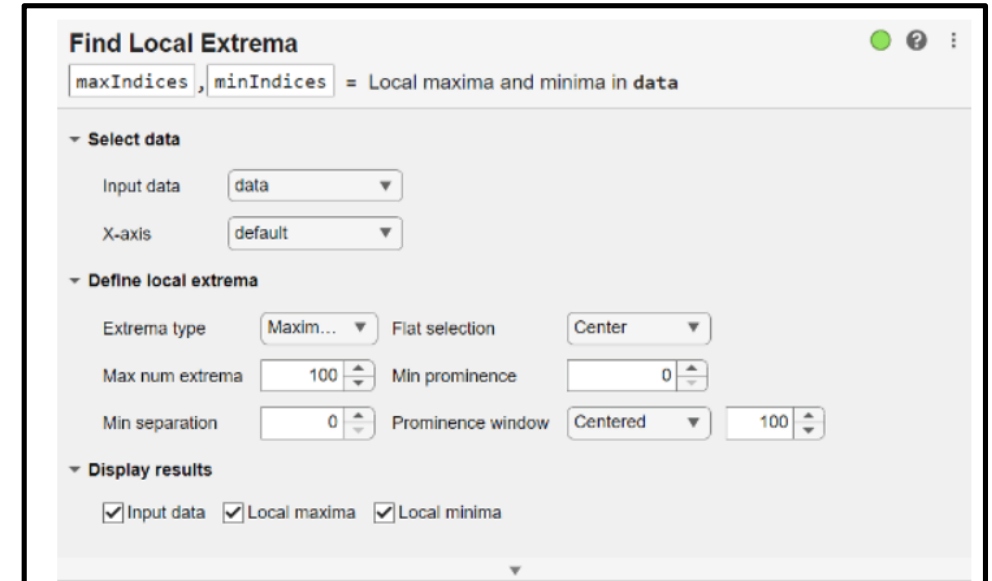
Since R2019b

Description

The **Find Local Extrema** task lets you interactively find local maxima and local minima in data. The task automatically generates MATLAB[®] code for your live script.

Using this task, you can:

- Find local maxima, minima, or both in data from a workspace variable.
- Adjust parameters to locate fewer or more extrema.
- Visualize the detected extrema.



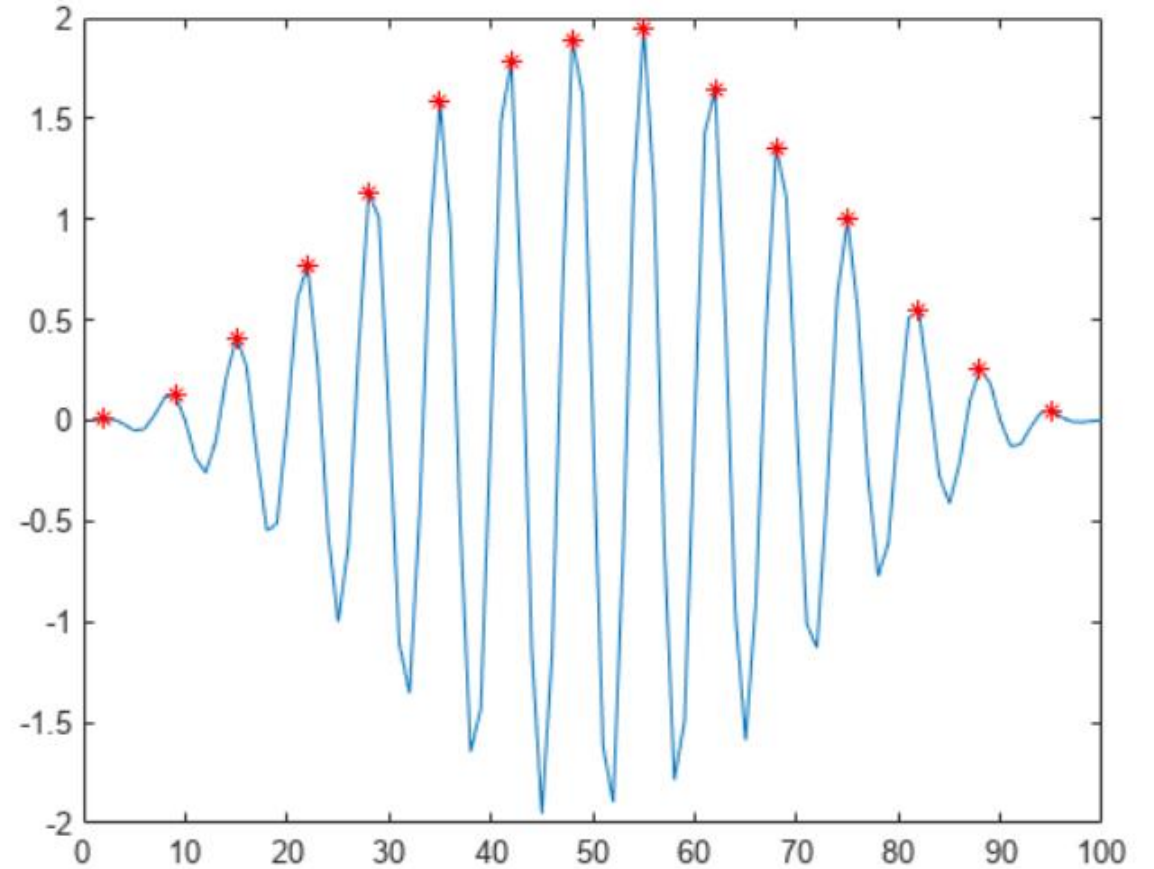
Other functions in MATLAB

islocalmax

Find local maxima

Syntax

```
TF = islocalmax(A)
TF = islocalmax(A,dim)
TF = islocalmax(__,Name,Value)
[TF,P] = islocalmax(__)
```



<https://www.mathworks.com/help/matlab/ref/islocalmax.html>

Other functions in MATLAB

findpeaks

Find local maxima

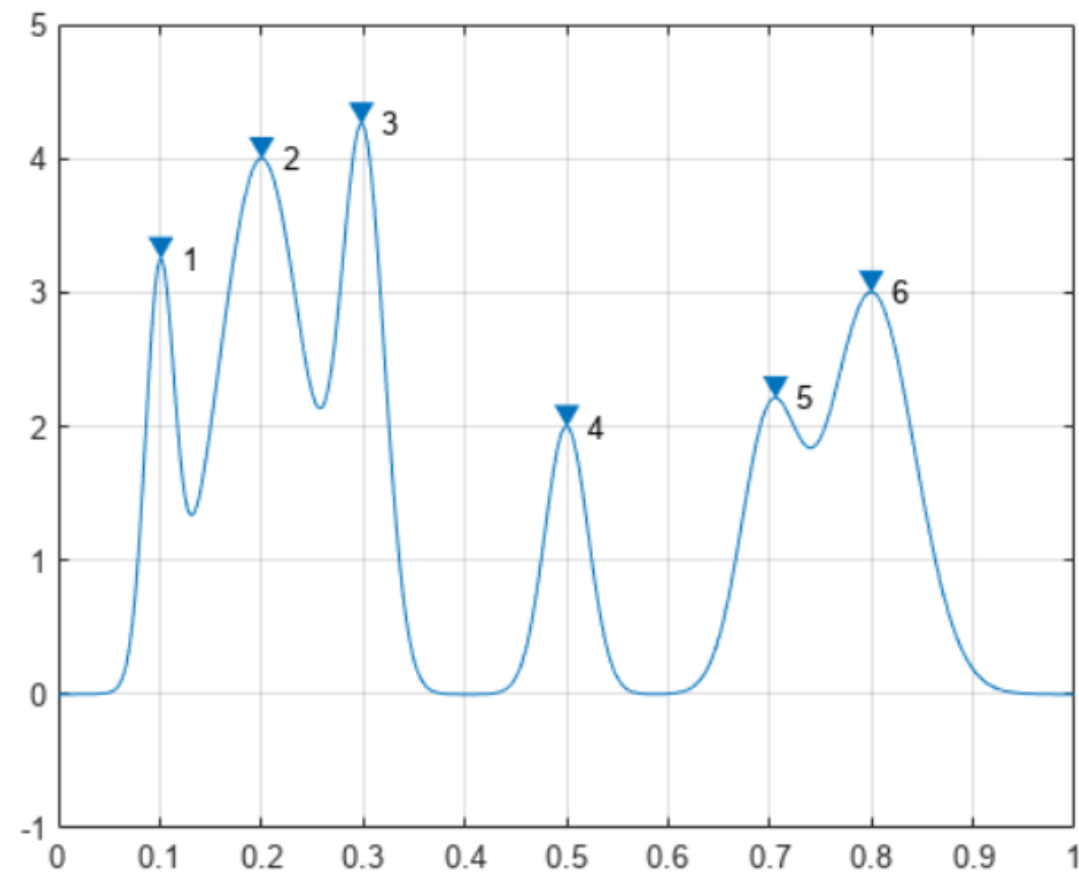
Syntax

```
pks = findpeaks(data)  
[pks,locs] = findpeaks(data)  
[pks,locs,w,p] = findpeaks(data)
```

```
[__] = findpeaks(data,x)  
[__] = findpeaks(data,Fs)
```

```
[__] = findpeaks(__,Name,Value)
```

```
findpeaks(__)
```



References

- [1] Apolinário, Isabela F., and Paulo SR Diniz. "Introduction to signal processing theory." In *Academic press library in signal processing*, vol. 1, pp. 3-28. Elsevier, 2014.